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Unveiling AI washing: Bridging corporate technological gaps through a cognitive dissonance lens☆

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ABSTRACT

This study utilizes Cognitive Dissonance Theory to empirically investigate how 'AI washing', the discrepancy between AI narratives and actual capabilities, affects the corporate technological gap. Using panel data from China's A-share listed firms (2007–2022), the findings establish a significant inverted U-shaped relationship between 'AI washing' and the technological gap. Mediation analysis confirms this relationship is channeled through both internal R&D investment and industry-level R&D investment. Moderation analysis reveals that strong AI-enabled participatory learning capability flattens the inverted U-curve, indicating earlier corrective action. Conversely, high investor sentiment is shown to steepen the curve. Furthermore, the nonlinear effect is subdued for firms in national AI pilot zones or high-technology-intensive industries. This research advances 'AI washing' literature through quantitative analysis, extends Cognitive Dissonance Theory to the domain of technology strategy, and offers empirical insights for responsible AI governance.

1. Introduction

Artificial intelligence (AI), characterized as a General-Purpose Technology (GPT) (Bahoo et al., 2023), has recently spurred significant advancements in corporate operational efficiency and innovation capability. However, this wave has been accompanied by a phenomenon termed 'AI washing,' wherein some firms mislead stakeholders by exaggerating AI capabilities or fabricating application scenarios (Bietti, 2020; Delmas and Burbano, 2011).

Existing research has explored the drivers of AI washing from perspectives such as technological characteristics and regulatory gaps (Bietti, 2020) and has partially investigated its impact on organizational legitimacy (Elsayed, 2025), investor confidence (Basnet et al., 2025), and corporate valuation (Xing et al., 2026). While there is general agreement that AI washing distorts resource allocation, the further implications of this resource misallocation remain underexplored. We posit that AI washing is likely to alter firms' innovation trajectories by misallocating resources and distorting market expectations, thereby fundamentally influencing technological gaps between firms. The research attention to this crucial connection remains insufficient.

Under the new wave of competition driven by AI technology, firms recognize the importance of AI innovation in narrowing technological gaps; however, the high costs and inherent uncertainties associated with AI R&D often impede the fulfilment of strategic AI commitments (Shi, 2024). This inherent misalignment between public narrative and internal action readily triggers cognitive dissonance at the organizational level, subsequently reshaping firm behavior and economic outcomes. Cognitive Dissonance Theory (Auster, 1965) focuses on the psychological tension arising from inconsistency between organizational beliefs (or statements) and actions, and the resultant behavioral adjustments driven by this tension. This theoretical logic is ideally suited for analyzing corporate conduct in the context of AI washing, providing an

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appropriate framework for understanding the resultant relationship with technological gaps, along with its transmission mechanisms and boundary conditions.

As the world's largest emerging economy, China presents a unique real-world context for studying AI washing, exhibiting both significant AI technological potential and evolving institutional settings. Using panel data from Chinese A-share listed firms spanning the period 2007 to 2022, our results reveal a significant inverted U-shaped relationship between AI washing and the corporate technological gap. Specifically, a low level of AI washing tends to widen the technological gap, whereas once the intensity surpasses a critical threshold, it begins to narrow the gap. Mechanism analysis indicates this effect operates through two distinct pathways: firm-level R&D investment and industry-level R&D investment. Furthermore, firms with stronger AI-enabled participatory learning capabilities exhibit a flatter inverted U-shaped curve, which facilitates an earlier behavioral adjustment. Heterogeneity analysis further demonstrates that the shape of the inverted U-shaped curve and the critical threshold differ significantly under varying conditions, including high investor sentiment, elevated R&D intensity, and stringent regulatory environments.

This study contributes to the existing literature in three key aspects. First, while prior research on AI washing has largely remained conceptual (Seele and Schultz, 2022), this paper draws on quantification methods to construct a measurable indicator for AI washing, thereby advancing the field toward rigorous empirical analysis. Second, moving beyond the conventional focus of cognitive dissonance theory on individual psychology, we extend its application to the domain of corporate strategic misrepresentation. By explicating how the conflict between narrative and action in AI washing influences the technological gap through cognitive tension, this research significantly broadens the theoretical perspective of cognitive dissonance. Third, addressing the static analytical limitations prevalent in existing studies on the technological gap (Burger et al., 2022; Fuenfschilling and Truffer, 2014), this study employs Cognitive Dissonance Theory to reveal the technological gap as a dynamic evolutionary process driven by cognitive dissonance, thereby enriching the literature on technological gap dynamics.

2. Literature review and theory development

2.1. AI washing

AI washing is a strategic misleading behavior. Similar to greenwashing (Delmas and Burbano, 2011), AI washing specifically refers to firms' misleading technological marketing in AI-related strategies, where they mislead stakeholders by exaggerating their AI capabilities or fabricating AI application scenarios (Seele and Schultz, 2022). In essence, the core purpose of AI washing is to leverage information asymmetry in technological narratives to gain external attention and resources, rather than commit substantial effort to R&D and application of AI technologies. This behavior typically endows firms with greater legitimacy and enables them to establish a better corporate image (Elsayed, 2025).

The emergence of AI washing results from the interplay between technological characteristics and institutional deficiencies (Bietti, 2020). From a technological perspective, AI's inherent attributes create opportune conditions for such behavior. On the one hand, its long R&D cycles and high uncertainty (Shi, 2024) impede firms from achieving rapid, substantial breakthroughs. On the other hand, AI's complexity and algorithmic opacity (Theodorou and Dignum, 2020) make it difficult for external stakeholders, such as ordinary investors and consumers, to accurately verify a firm's claimed AI capabilities (Seele and Schultz, 2022). These factors collectively create space for misleading disclosures through exaggerated narratives (Adomako and Tran, 2023). From an institutional perspective, the current underdeveloped governance framework for AI widens the arbitrage space between legal risk and potential return for AI washing (Jobin et al., 2019), which ultimately incentivizes this corporate behavior.

AI washing has a profound impact on the social system and organizational change. In the short term, AI washing may create the illusion of AI technology prosperity, attract capital into the AI field, and boost the market expectation of relevant industries (Bietti, 2020). Additionally, compelling AI narratives can capture the attention of stakeholders and increase their acceptance and participation in AI technology (Denia, 2025), potentially enhancing corporate value (Xing et al., 2026). Conversely, firms that abstain from engaging in proactive AI narratives may suffer a loss of investment sentiment (Basnet et al., 2025). However, in the long run, widespread AI washing behavior may trigger a systematic crisis of trust in AI technology and its development path. Simultaneously, it risks simplifying genuine innovation into mere marketing rhetoric, distorting social values, and hindering the establishment and improvement of the AI governance ethics framework and relevant laws and regulations (Seele and Schultz, 2022).

To summarize, existing studies have successfully delineated the concept, driving factors, and potential impacts of AI washing. Table 1 presents the key existing literature related to AI washing. However, most extant work relies on descriptive analysis, with a notable lack of empirical research based on quantitative measurement tools. This makes it difficult to accurately capture the intensity of AI washing and its specific, measurable impact on corporate technology strategy. To address this gap, this study constructs a novel quantitative AI washing index and empirically explores its dynamic relationship with corporate technological gaps.

Table 1
Key literature related to AI washing.

Research theme	Key aspects	Literature and corresponding argument
Concept of AI washing	Misleading behavior	Seele and Schultz (2022) clearly define AI washing as a practice in which firms mislead stakeholders by overstating AI capabilities or fabricating application scenarios.
	Exaggerated narrative	Xing et al. (2026) characterizes it as a form of exaggerated corporate AI narrative.
Driving factor	Investment risks	Shi (2024) underscores the extended R&D cycles and high uncertainty associated with AI technology.
	Technological characteristics	Martin (2019) and Theodorou and Dignum (2020) emphasize the inherent complexity and algorithmic opacity of AI, which, as noted by Seele and Schultz (2022) and Martin (2019), makes it challenging for external stakeholders to verify a firm's claimed AI capabilities.
	Inadequate regulation	Jobin (2019) points out that current laws and regulations in the field of AI remain ambiguous and regulatory mechanisms are often ineffective.
Impact	Short-term impact	Bietti (2020) suggests that AI washing can create a short-term illusion of technological progress, attract capital inflows, and raise market expectations. Denia (2025) observes that AI narratives can capture stakeholder attention and increase acceptance of and engagement with AI technology. Xing et al. (2026) argues that AI narratives can enhance corporate value.
	Long-term impact	Seele and Schultz (2022) warn that AI washing may trigger a crisis of trust, distort social values, and hinder the establishment and improvement of AI governance frameworks along with related laws and regulations.

2.2. Cognitive dissonance theory

Cognitive dissonance theory, initially proposed by Festinger (1957), explains the psychological discomfort that individuals or organizations experience when confronted with conflicting cognitions. When discrepancies arise between different cognitions or between cognition and behavior, it will trigger psychological tension (Scher and Cooper, 1989) and urge actors to restore the consistency between their cognition and behavior (Pugh et al., 2011). This tension is a psychological discomfort, as well as a significant driver of behavioral adjustment (Pugh et al., 2011). Specifically, when the degree of dissonance exceeds an individual's tolerance threshold for inconsistency, two main coping mechanisms are usually used: one involves changing behavior to keep consistent with existing cognition, and the other involves modifying cognition to rationalize existing behavior.

Although this theory has been widely applied at the individual level to explain consumer decision-making (Powers and Jack, 2013) and financial investment behavior (Li et al., 2023), recent studies have extended its application to the organizational level, which provides insights on how organizations respond to dissonance during strategic implementation (Kim and Bay, 2017). In this context, cognitive dissonance is manifested in the inconsistency between organizational beliefs, strategic goals and practices (Rabin, 1994). Such dissonance may have far-reaching negative consequences, including reputation damage (Kim and Bay, 2017) and failure of open innovation programs (Bhimani et al., 2023). For example, prior research has applied cognitive dissonance theory to examine the tension between firms' commitments to sustainable practices and their pursuit of formal certification (Flagstad et al., 2022), which confirms the argument that organizational dissonance can trigger behavior or strategic change.

In the context of AI washing, it has been witnessed that many firms publicly promote ambitious AI strategies and technological visions but lack internal capabilities or actual innovation capabilities to confirm these claims. This constitutes a typical case of organizational cognitive dissonance for firms. In the environment of high public concern and increasing regulatory pressure, the tension caused by such dissonance has further amplified, prompting firms to take corrective actions to re-establish the consistency between cognition and behavior (Rabin, 1994). Thus, cognitive dissonance theory not only offers a psychological explanation for AI washing behavior but also provides a theoretical foundation for understanding how the behavior adjustment induced by dissonance affects the technological trajectory and technological gaps of firms. This approach helps to expand the application boundary of cognitive dissonance theory in the field of organizational management and strategic research.

2.3. AI washing and corporate technological gap

According to the theory of cognitive dissonance, the inconsistency between cognition and behavior will produce internal psychological tension (Rabin, 1994). In the context of AI washing, firms exaggerate their AI capabilities to establish the image of technological pioneers, to obtain short-term benefits such as market recognition and policy support.

At low levels of AI washing, as mentioned above, these instant benefits attracted by AI washing provide psychological returns for firms, enabling them to rationalize the inconsistency between their claimed AI innovation commitment and actual technology practice (Seele and Schultz, 2022), and firms will not have a strong sense of crisis. Nevertheless, this illusion of cognitive stability will subtly lead to the technological gap between these AI washing firms and those competitors with authentic AI capabilities (Sun et al., 2025a, 2025b), especially the industry leaders.

However, cognitive dissonance theory also posits that rationalization mechanism has limitations. With the intensification of AI washing, the contradiction between the understanding of the strategic importance of AI within firms and the actual lack of R&D investment has become increasingly prominent. When AI washing behavior reaches the critical threshold, this tension reaches its peak, and the technological gap between the focal firm and industry leaders also tends to reach its maximum. In addition, the accumulated dissonance exceeds the psychological tolerance of the firms (Pugh et al., 2011), leading to the collapse of the existing rationalization mechanism. This triggers a process of behavior adjustment, which encouraged firms to launch substantial technological efforts to restore the consistency between their strategic cognition and organizational action. These efforts can be shown as increasing R&D investment and genuine push for technological innovation, to effectively narrow the technological gap.

For example, in China's AI industry, some firms, such as iFLYTEK, frequently publicized AI related concepts to strengthen their corporate image in the early stage, but due to the limited actual technology reserves, the technological gap between them and industry leaders has widened in the short term. With the deepening of AI washing, there is a significant cognitive dissonance between high profile AI image and technological reality among the internal members of firms. Driven by internal psychological pressure, firms tend to alleviate this dissonance by increasing R&D investment, optimizing technology strategies and introducing professionals, so as to gradually narrow the technological gap. Based on the above reasoning, we propose the following hypothesis:

H1. There is an inverted U-shaped relationship between AI washing and the corporate technological gap, such that the gap initially widens and then narrows as the level of AI washing increases.

2.4. The mediating roles of firm-level and industry-level R&D investment

The technological gap of firms is affected not only by internal technological evolution within firms, but also by the overall dynamic changes of the industrial technology ecosystem. Therefore, this study explores the mechanism through which AI washing affects corporate technological gaps from both firm and industry levels.

When the degree of AI washing is low, the firm-level R&D investment tends to remain stable. However, with the intensification of AI washing, that is, firms gradually shift their focus from substantive innovation to symbolic communication, this means that firms will allocate more resources to technological marketing activities rather than actual R&D investment. This shift in resource allocation usually triggers a “crowding out” effect, where expenditures on symbolic expression replaces R&D investment (Sun et al., 2025a, 2025b). With the reduction of R&D investment, firms' technological innovation capabilities become increasingly constrained (Sun et al., 2025a, 2025b), further widening the gap with the industry's technological frontier.

When the degree of AI washing exceeds a certain threshold, the psychological tension caused by organizational cognitive dissonance will break through the tolerance boundary of the firm. This is because when a firm shapes the image of technological leadership through AI washing, while the contradiction with its actual backward technological capabilities accumulates to a certain extent, the internal management team and employees will have fierce psychological dissonance. At this point, the consistency restoration mechanism emphasized by cognitive dissonance theory will be activated (Pugh et al., 2011). To restore the consistency between cognition and behavior and alleviate the discomfort caused by the mismatch between the claimed capabilities and the actual capabilities, firms will launch substantial technological investment. Increasing R&D investment and promoting genuine technological innovation have become the keys to resolving this conflict (Bhimani et al., 2023). In addition, the reputation gains and resource accumulation brought by AI washing in the short term (Bietti, 2020) also provide necessary financial foundation for the R&D investment rebound of firms, which enables firms to invest the benefits obtained from symbolic behavior into substantive innovation activities.

To sum up, there is a U-shaped relationship between AI washing and

firm-level R&D investment. As the degree of AI washing increases, R&D investment first decreases and then rebounds. This pattern has further contributed to the inverted U-shaped relationship between AI washing and corporate technological gaps, that is, to widen the gap first, and then gradually narrow it. Based on the above analysis, we propose the following hypothesis:

H2. AI washing has a U-shaped relationship with firm-level R&D investment. Specifically, as the degree of AI washing intensifies, firm-level R&D investment first decreases and then increases, which leads to the technological gap being first widened and then narrowed.

Meanwhile, when a firm's AI washing behavior remains at a low level, it is typically insufficient to attract significant attention from peer firms or the market. Therefore, the R&D investment at the industry level tends to be stable. However, with the increase of AI washing, two key mechanisms begin to appear.

First, peer firms in the same industry may begin to compare themselves with the so-called AI pioneer firms. This benchmarking often generates a mismatch between the peers' current capabilities and the so-called technological standards set by the AI washed firms, resulting in cognitive dissonance within the industry. To resolve this dissonance, peer firms may reallocate their resources to enhance their own AI capacities. This may lead to a surge in R&D investment in the entire industry, which is similar to the R&D arms race (Carrier and Tushnet, 2021).

Second, investors usually evaluate the entire industry by referring to the successful experience of such “AI pioneer” firms. As these firms expand their AI-related narratives, investors may also experience the dissonance between their existing perception regarding the industry and the technological frontier “owned” by the AI washed firms. To reconcile this discrepancy, investors may increase investment in the industry, thus indirectly promoting industry-level R&D investment. This increase in R&D investment pushes the industry's real technological frontier forward. However, because the “AI pioneer” firms have not actually improved its genuine AI capabilities, they have failed to keep up with the real progress of the industry, thereby widening the gap between firms and the industry's technological frontier.

Once AI washing exceeds a certain critical threshold, the credibility of the firm's technological marketing begins to decline. For peer firms, the psychological tension induced by cognitive dissonance will be weakened, because AI washing behavior is no longer considered to reflect the actual AI capability (Seele and Schultz, 2022). Consequently, peer firms reduce or cease imitating the efforts of AI washed firms, and the industry-level R&D investment tends to be stable. In contrast, industry investors may interpret the heightened AI washing as a signal of bubble risk, which will lead to the decline of investment in the whole industry. With the reduction of available R&D investment, the speed of technological progress in the whole industry slows down, thereby narrowing the gap between the AI washed firm and the industry leader.

In sum, the intensification of AI washing gives rise to a nonlinear, inverted U-shaped trajectory of industry-level R&D investment, that is, R&D investment at the industry level will first increase and then decline. This dynamic change plays a key role in driving the inverted U-shaped relationship between AI washing and corporate technological gap. Based on this analysis, we propose the following hypothesis:

H3. AI washing has a U-shaped relationship with industry-level R&D investment. Specifically, as the degree of AI washing intensifies, industry-level R&D investment first increases and then decreases, which leads to the technological gap first widened and then narrowed.

2.5. The moderating role of AI-enabled participatory learning capability

AI-enabled participatory learning is an emerging organizational learning mechanism, which combines AI with human cognitive capability (Falco, 2019). It emphasizes human–machine collaboration in the organizational process to enhance the efficiency of knowledge acquisition, technological absorption and innovation (Falco, 2019). Through confrontational design, discourse interaction and iterative feedback, AI-enabled participatory learning can be able to expand the knowledge boundary of individuals and organizations, and provide decision-makers with different perspectives and ways of problem-solving (Wilson et al., 2025). It acts as a cognitive bridge, which links the internal organizational learning system with the external technological ecosystem (Zirar et al., 2023), thus plays a key role in facilitating intelligent transformation and strategic alignment.

In the context of AI washing, AI-enabled participatory learning helps firms identify the risks associated with the dissonance in an earlier stage, to accelerate the transformation from technology marketing to substantive investment through human–machine collaborative learning. Specifically, a high-level AI-enabled participatory learning improves the sensitivity of both management and employees to cognitive biases induced by AI washing. This is because the participatory learning supported by AI significantly improves the firm's information processing capacity and the ability to identify technological trends. Through in-depth interaction with AI systems, firms can generate novel problem solutions, facilitate resource restructuring, and accelerate strategic technology deployment (Wang et al., 2024). In this process, AI serves not only as a cognitive support tool, but also as a behavior guidance mechanism, enabling firms to transform the dissonance triggered by AI washing into a driving force for genuine innovation (Liu et al., 2025). Moreover, AI-enabled participatory learning allows firms to more accurately evaluate the technological gap between themselves and the industry frontier (Saemundsson and Candi, 2017), to promote earlier behavior adjustments, boost R&D investment, and ultimately narrow the technological gap.

The improved cognitive and decision-making sensitivity resulting from AI-enabled participatory learning leads to a flattening of the inverted U-shaped relationship between ‘AI washing’ and the technological gap. Under conditions of high participatory learning, firms tend to initiate corrective behavioral adjustments before the accumulated cognitive dissonance reaches the critical threshold, thereby mitigating the potential negative consequences of ‘AI washing’. Conversely, a deficiency in AI-enabled participatory learning capability increases the likelihood that firms will be misled by the strategic narrative, delaying the necessary behavioral correction and leading to a more pronounced widening of the technological gap (Saemundsson and Candi, 2017).

In summary, AI-enabled participatory learning mitigates the inverted U-shaped relationship between AI washing and the technological gap by enhancing firms' ability to recognize and adapt to cognitive dissonance. Under high levels of AI-enabled participatory learning, the curve becomes flatter, and the negative consequences of AI washing are significantly reduced.

H4. AI-enabled participatory learning capability moderates the inverted U-shaped relationship between AI washing and the technological gap, that is, the relationship flattens under higher levels of participatory learning.

3. Research design

3.1. Data sources and sample selection

This study constructs a multi-source panel dataset of Chinese A-share listed firms covering the period from 2007 to 2022, aiming to investigate the impact of AI washing behavior on firms' technological gaps. Specifically, AI-related disclosure data are obtained from firms' annual reports; patent data are sourced from the China National Intellectual Property Administration (CNIPA); financial data come from the China Stock Market and Accounting Research (CSMAR) database; and media coverage data are collected from the China Financial and Economic News Database (CFDN). These data are merged by the stock codes of

firms and years to form an unbalanced firm-year panel dataset. The year 2007 is chosen as the starting point since AI-related information in annual reports prior to this year was extremely limited, rendering measurement infeasible. To ensure the robustness of the sample, we adopted several screening criteria. (1) According to the industry classification standard of the National Bureau of Statistics of China, firms in the financial industry are excluded; (2) Firms designated as ST, *ST, or PT are excluded because potential financial anomalies may affect the disclosure behavior; (3) Observations with missing values are removed; And (4) all continuous variables are winsorized at the 1st and 99th percentiles to mitigate the influence of outliers. The final sample comprises 3842 listed firms in 17 industries, with a total of 36,200 firm-year observations. Table 2 reports the sample distribution in detail.

3.2. Variable measurement

3.2.1. Independent variable

The key independent variable in this study is the degree of corporate AI washing. Drawing upon and extending the conceptual and methodological frameworks commonly used to measure greenwashing behaviors (Zhang, 2022, 2023), we constructed the AI washing index based on the discrepancy between the level of corporate disclosure regarding AI technologies and the actual technological capabilities. Specifically, for AI-related disclosure, we develop a proprietary AI keyword dictionary (see Appendix Table A1) and utilize web scraping techniques to extract the frequency of AI-related keywords appearing in annual reports. Annual reports are a crucial channel through which firms communicate their strategic intentions and technological commitments to stakeholders. As such, the frequency of AI keyword usage is widely recognized as a proxy for a firm's publicly declared AI strategy and commitment (Bai et al., 2024).

In terms of a firm's actual AI capabilities, it is measured by the number of AI-related patents granted each year, which reflects the firm's real R&D outputs and accumulated expertise in the AI domain. This metric is broadly used to assess technological innovation capacity. To mitigate comparability biases arising from industry heterogeneity and firm size variation, we standardize both the disclosure frequency and the patent counts. The degree of AI washing is then calculated using the following formula:

$$AIwashing_{it} = \frac{AI_{disclosure\ it} - \overline{AI_{disclosure\ t}}}{\sigma AI_{disclosure}} - \frac{AI_{patent\ it} - \overline{AI_{patent\ t}}}{\sigma AI_{patent}} \quad (1)$$

whereas $AIwashing_{it}$ denotes the degree of AI washing for firm i in year t, $AI_{disclosure\ it}$ represents the frequency of AI-related keywords disclosed in firm i's annual report in year t, and $AI_{patent\ it}$ indicates the number of AI-related patents granted to firm i in year t. $\overline{AI_{disclosure\ t}}$ and $\overline{AI_{patent\ t}}$ denote the cross-sectional averages of AI keyword frequency and AI patent grants for all firms in year t, respectively. $\sigma AI_{disclosure}$ and σAI_{patent} refer to the standard deviations of the respective variables across all firms in year t. This index captures the extent of misalignment between a firm's AI-related publicity and its actual technological capacity. A larger deviation indicates a higher level of AI washing, suggesting that the firm overstates its AI capabilities relative to its real technological achievements.

3.2.2. Dependent variable

The technological gap typically refers to the systematic differences among firms in terms of knowledge base, technological capability, and managerial practices (Bourles et al., 2013). A commonly used method in existing literature measures this gap by comparing a firm's total factor productivity (TFP) to that of the productivity frontier—usually defined as the firm with the highest TFP within the same industry (Bourles et al., 2013). However, this approach primarily captures disparities in production efficiency and does not directly reflect differences in technological output. Moreover, methods such as Data Envelopment Analysis (DEA), frequently used to estimate the technological frontier, are sensitive to data stability and lack a standardized measurement scale (Ali and Lerme, 1997).

To address these limitations, this study adopts patent output differentials as an alternative proxy for measuring a firm's technological gap, following the approach of Liang et al. (2022). Specifically, we use the total number of patents granted to a firm each year as an indicator of its technological innovation output. The technological frontier within each industry is defined as the firm with the highest number of patent grants in that year. The technological gap is then calculated using the following formula:

$$\mathrm{TechGap}_{it} = \ln\left(\mathrm{Patent}^{h}_{\mathrm{frontier,t}} - \mathrm{Patent}_{it}\right) \quad (2)$$

where $\mathrm{TechGap}_{it}$ denotes the technological gap of firm i in year t, $\mathrm{Patent}^{h}_{\mathrm{frontier,t}}$ is the number of patents granted to the firm with the highest patent grants in industry h during year t, and Patent_{it} is the total number of patents granted to firm i in the same year.

3.2.3. Mediating and moderating variables

As proposed in the previous hypotheses, AI washing behavior may indirectly affect a firm's technological gap by influencing both its own R&D investment and that of its industry. Accordingly, this study introduces two mediating variables: firm-level R&D investment, measured by the natural logarithm of the total annual R&D expenditure of the firm (Sheng and Ni, 2024); and industry-level R&D investment, measured by the natural logarithm of the average R&D expenditure of all firms within the same industry each year (Coad, 2019).

This study introduces AI-enabled participatory learning capability as a moderating variable to capture a firm's internal capacity to absorb and apply AI technologies. Drawing on existing literature (Wang et al., 2024), learning capability is commonly proxied by R&D intensity (i.e., the ratio of R&D expenditure to total assets). However, since firms do not separately disclose AI-specific R&D investment in their annual reports, this study estimates the variable using patent data from the CNIPA. Specifically, we first calculated the proportion of AI-related granted patents to total granted patents for each firm each year, which

Table 2
Sample distribution.

Industry code	Industry category	Number of firms	Number of observations
A	Agriculture, Forestry, Animal Husbandry, and Fishery	41	456
B	Mining Industry	75	923
C	Manufacturing	2492	23,410
D	Electricity, Heat, Gas, and Water Production and Supply	122	1405
E	Construction	88	865
F	Wholesale and Retail Trade	164	1826
G	Transportation, Storage, and Postal Services	105	1189
H	Accommodation and Catering Services	6	35
I	Information Transmission, Software, and Information Services	345	2583
K	Real Estate	94	1315
L	Leasing and Business Services	60	566
M	Scientific Research and Technical Services	84	400
N	Water Conservancy, Environment and Public Facilities Management	79	602
P	Education	7	51
Q	Health and Social Work	13	98
R	Culture, Sports, and Entertainment	56	514
S	Public Administration, Social Security, and Social Organizations	11	160
Total		3842	36,200

reflects the share of AI-related technologies in the firm's overall R&D output. Then, we multiply this ratio by the firm's total R&D expenditure per employee (i.e., total R&D spending divided by the number of employees) to obtain an index of AI participatory learning capability. A higher value indicates that more human resources are devoted to AI-related R&D, increasing employees' exposure to and engagement with AI technologies, and thus implying a higher level of participatory learning. The level of AI-enabled participatory learning capability is then calculated using the following formula:

$$AI-enabled\ participatory\ learning\ capability_{it} = \frac{AI_{patent\ it}}{Patent_{it}} \times \frac{R\&D\ expenditure_{it}}{Employee\ number_{it}} \tag{3}$$

3.2.4. Control variables

Following prior studies (Xu et al., 2024), this paper includes a set of control variables based on firms' fundamental characteristics and corporate governance structure. For details, see Table 3. Regarding firm characteristics, we control firm size using the natural logarithm of total assets; leverage is measured as the ratio of total liabilities to total assets at year-end; firm listing age is calculated as the natural logarithm of the difference between the current year and the IPO year plus one. Additionally, the fixed asset ratio is defined as the ratio of net fixed assets to total assets; board size is represented by the natural logarithm of the number of directors; and return on assets is computed as net profit divided by the average total assets.

In terms of corporate governance, we consider the effect of executive incentive mechanisms. Drawing on Xu et al. (2024), executive ownership percentage is measured as the ratio of the number of shares held by directors, supervisors, and senior executives to the firm's total shares outstanding, while executive compensation is captured by the natural logarithm of the total compensation paid to top executives. Furthermore, to reflect the degree of ownership concentration, we use ownership balance, measured as the ratio of the combined shareholding of the second to fifth largest shareholders to that of the largest shareholder (Wang and Cheng, 2023).

Table 3
Variable measurement.

Variable type	Variable name	Measurement
Independent variable	AI washing	See Formula (1)
Dependent variable	Technological gap	See Formula (2)
Mediating variables	Firm-level R&D investment	Natural logarithm of annual R&D expenditure
	Industry-level R&D investment	Natural logarithm of average industry R&D expenditure
Moderating variable	AI-enabled participatory learning capability	See Formula (3)
Control variables	Return on assets	Net profit divided by average total assets
	Fixed asset ratio	Ratio of fixed net assets to total assets
	Executive compensation	Natural logarithm of total executive compensation
	Executive ownership percentage	Shareholding ratio of executives to total equity
	Board size	Natural logarithm of the number of board members
	Ownership balance	Shareholding of 2nd–5th largest shareholders divided by that of the largest
	Listing age	Logarithm of (current year – listing year +1)
	Firm size	Natural logarithms of total assets
	Leverage	Ratio of total liabilities to total assets at year-end

3.3. Research design

To examine the impact of AI washing on technological gaps among listed firms, this study constructs a two-way fixed effects model as follows:

$$Technological\ gap_{it} = \beta_0 + \beta_1 AIwashing_{it} + \beta_2 AIwashing_{it}^2 + \sum \beta\ Controls_{it} + Year + Industry + \varepsilon_{it} \tag{4}$$

where Technological gap_{it} denotes the technological gap of firm i in year t; $AIwashing_{it}$ and $AIwashing_{it}^2$ represent the level and squared term of AI washing, respectively; $Controls_{it}$ is a vector of control variables; Year and Industry denote year and industry fixed effects, respectively, which account for unobservable temporal trends and industry-specific heterogeneity; and ε_{it} is the error term. Robust standard errors are used in model estimation.

Furthermore, we extend the analysis by constructing models for the mediating and moderating effects. The selected control variables and fixed effects remain consistent with the baseline regression model. The mediating effect model is specified as:

$$Firm\ (or\ Industry)\ level\ R\&D\ investment_{it} = \beta_0 + \beta_1 AIwashing_{it} + \beta_2 AIwashing_{it}^2 + \sum \beta\ Controls_{it} + Year + Industry + \varepsilon_{it} \tag{5}$$

$$Technological\ gap_{it} = \beta_0 + \beta_1 AIwashing_{it} + \beta_2 AIwashing_{it}^2 + Firm\ (or\ Industry)\ level\ R\&D\ investment_{it} + \sum \beta\ Controls_{it} + Year + Industry + \varepsilon_{it} \tag{6}$$

To test the moderating effect of AI-enabled participatory learning capability, we construct the following moderating effect model:

$$Technological\ gap_{it} = \beta_0 + \beta_1 AI\ enabled\ participatory\ learning\ capability \times AIwashing_{it} + \beta_2 AI\ enabled\ participatory\ learning\ capability \times AIwashing_{it}^2 + \beta_3 AI\ enabled\ participatory\ learning\ capability + \sum \beta\ Controls_{it} + Year + Industry + \varepsilon_{it} \tag{7}$$

4. Empirical results

4.1. Descriptive statistics and correlation analysis

This study begins with descriptive statistics and correlation analysis. Table 4 shows that the mean value of technological gap is 5.912, and the standard deviation is 2.511, indicating significant heterogeneity in terms of technological gaps among firms. Compared with the industry leader, some firms show a large gap (the maximum value is 8.034). The mean of AI washing is 0.019, and the standard deviation is 0.769, suggesting that most firms maintain a relatively low level of AI washing. However, the wide range of values (from −2.413 to 4.469) reveals that some firms have greatly exaggerated their AI-related disclosures. The results of other variables are consistent with previous literature. The Pearson correlation coefficients of all variables are below 0.514, suggesting no obvious multicollinearity. To further exclude the possibility of multicollinearity, the variance inflation factor (VIF) of each variable is calculated. All VIF values are below 2 (with a maximum of 1.96), which is far below the conventional threshold of 10, indicating that multicollinearity is not a serious concern in this study.

4.2. Baseline regression results

To examine whether AI washing behavior significantly affects firms' technological gaps, this study first conducts a regression analysis

Table 4
Correlation analysis and descriptive statistics.

	1	2	3	4	5	6	7	8	9	10	Mean	Standard deviation	Minimum	Maximum	VIF	Observations
1. Technological gap	1.000										5.912	2.511	0	8.034		36,200
2. AI washing	−0.026***	1.000									0.019	0.769	−2.413	4.469	1.40	36,200
3. Return on assets	0.072***	−0.034***	1.000								0.037	0.064	−0.263	0.197	1.27	36,200
4. Fixed asset ratio	0.028***	−0.159***	−0.085***	1.000							0.221	0.164	0.002	0.708	1.11	36,200
5. Executive compensation	0.049***	−0.012**	0.197***	−0139***	1.000						15.260	0.795	13.155	17.309	1.64	36,200
6. Board size	−0.119***	−0.078***	0.016***		0.167***	1.000					2.134	0.200	1.609	2.708	1.10	36,200
7. Ownership balance	0.086***	0.073***	0.004		−0.099***	0.133***	−0.000	1.000			0.720	0.604	0.0264	2.769	10.7	36,200
8. Listing age	−0.010*	−0.071***	−0.090***		−0.034***	0.234***	−0.027***	0.019***	1.000		2.877	0.349	1.792	3.526	1.10	36,200
9. Firm size	−0.146***	−0.143***	0.027***		0.087***	0.514***	0.238***	−0.096***	0.206***	1.000	22.186	1.306	19.713	26.095	1.96	36,200
10. Leverage	−0.230***	−0.103***	−0.367***		0.117***	0.034***	0.154***	−0.129***	0.122***	0.430***	0.433	0.208	0.055	0.937	1.56	36,200

Note:***, **, and * denote significance levels at 1 %, 5 %, and 10 %, respectively.

Table 5
Baseline regression results.

Variables	(1)	(2)
AI washing	0.106***	0.104***
	(0.00440)	(0.00448)
AIwashing^2	−0.0197***	−0.0196***
	(0.000811)	(0.000825)
ROA		−0.280***
		(0.0494)
Fixed assets ratio		−0.0406**
		(0.0201)
Executive compensation		−0.0125***
		(0.00480)
Board size		0.00454
		(0.0149)
Ownership balance		−0.00476
		(0.00475)
Listing age		0.0121
		(0.00959)
Firm size		0.00395
		(0.00307)
Leverage		−0.0312*
		(0.0175)
Constant	5.926***	6.020***
	(0.00282)	(0.0748)
Year fixed effect	yes	yes
Industry fixed effect	yes	yes
Observations	36,249	36,200
R^2	0.957	0.957

Note: Robust standard errors are reported in parentheses. ***, **, and * denote significance at the 1 %, 5 %, and 10 % levels, respectively.

controlling only for year and industry fixed effects, without including any control variables. The results are presented in Column (1) of Table 5. The coefficient of the linear term of AI washing is positive (β_1=0.106, p < 0.01), and the coefficient of the quadratic term is negative (β_2= − 0.0197, p < 0.01). This suggests a potential nonlinear relationship between AI washing and corporate technological gap. Column (2) includes a full set of control variables. The regression results remain robust: the linear term of AI washing remains positive (β_1=0.104, $p < 0.01$), and the quadratic term remains negative (β_2= − 0.0196, p < 0.01). Using the standard turning point formula, the calculated inflection point is 2.6531. To further verify the existence of a nonlinear relationship, we perform a *U* test. The slope range is between −0.0718 and 0.1988, confirming the presence of an inverted U-shaped relationship between AI washing and technological gap.

From an economic perspective, moderate AI washing can help firms acquire external resources by shaping the technological image and catering to market expectations. However, due to the lack of actual resource investment, such behavior may widen the technological gap. With the intensification of AI washing, it may lead to resource misallocation and the decline of technological advantages. However, when AI washing becomes excessive, it means that AI-related disclosures deviate significantly from the actual AI capability of the firm, which may lead to strong cognitive dissonance. This in turn forces firms to increase R&D investment, enhance their AI capabilities, and ultimately narrow the technological gap. In sum, the effect of AI washing on technological gaps is not linear but follows an inverted U-shaped non-linear pattern, as illustrated in Fig. 1. Hypothesis H1 is thus supported.

4.3. Robustness checks

To verify the robustness of the baseline regression results, this study conducts a series of robustness tests from multiple dimensions, including (1) replacing the key explanatory and dependent variables, (2) adopting more stringent fixed effects specifications, and (3) controlling for potential omitted variables. The regression results are summarized in Table 6 and consistently support the existence of a robust inverted U-shaped relationship between AI washing and firms' technological gaps.

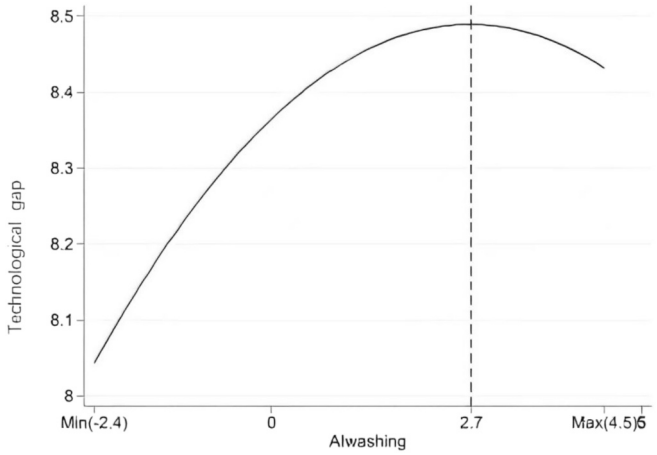


Fig. 1. Inverted U-shaped relationship between AI washing and firms' technological gaps.

4.3.1. Replacing the key independent variable

Given that AI-related keyword frequency in the full text of annual reports may be influenced by report layout and redundant phrasing, we use the frequency of AI-related keywords in the Management Discussion and Analysis (MD&A) section as a substitute. The MD&A section better reflects strategic priorities and managerial effectiveness and thus carries stronger signaling value. As shown in Column (1) of Table 6, the coefficients of the linear term (β_1=0.0405) and the quadratic term (β_2= − 0.00259) of AI washing remain significant at the 1 % level and are consistent with the baseline results, confirming the existence of an inverted U-shaped relationship.

4.3.2. Replacing the dependent variable

While the baseline regression uses the number of granted patents as a proxy for technological output, this may underestimate current innovation capability due to lengthy examination and approval procedures. As a robustness check, we replace granted patents with patent applications. Column (2) of Table 6 shows that the results still support the inverted U-shaped relationship (β_1=0.0442, $p < 0.01$; β_2= − 0.00529, p < 0.01), indicating that our findings are not sensitive to the choice of technological innovation output proxy.

4.3.3. Adopting more stringent fixed effects specifications

First, we replace industry fixed effects with firm fixed effects to account for unobservable time-invariant firm-level heterogeneity. Second, we include interaction fixed effects of industry and city and cluster

Table 6
Robustness and endogeneity test results.

Variables	(1) Replacing independent variable	(2) Replacing dependent variable	(3) Firm fixed effects	(4) Industry × city fixed effect + clustered standard error	(5) Controlling for omitted variables	(6) IV-first stage	(7) IV-second stage	(8) PSM
AI washing	0.0405***	0.0442***	0.0816***	0.103**	0.115***		0.356***	0.059***
	(0.00217)	(0.00386)	(0.00655)	(0.0385)	(0.00494)		(0.103)	(11.04)
AIwashing^2	−0.00259***	−0.00529***	−0.00627***	−0.0193***	−0.0204***		−0.0588***	−0.008***
	(0.0008)	(0.000725)	(0.00115)	(0.00521)	(0.000901)		(0.0102)	(−8.15)
ROA	−0.275***	−0.329***	−0.412***	−0.314	−0.146**	−3.364***	−0.224***	−0.302***
	(0.0487)	(−0.0418)	(−0.0543)	(−0.262)	(−0.0617)	(0.379)	(0.0808)	(5.65)
Fixed assets ratio	−0.0616***	−0.0387**	−0.118***	−0.0722	−0.0509**	−2.137***	−0.0315	−0.057***
	(0.0197)	(−0.017)	(−0.0339)	(0.116)	(0.0218)	(−0.147)	(0.0408)	(−2.60)
Executive compensation	−0.0111**	−0.0185***	−0.0373***	−0.00541	−0.0129**	−0.00576***	−0.011	−0.001
	(0.00471)	(0.00405)	(−0.00742)	(0.0251)	(0.00527)	(0.00161)	(−0.00672)	(0.25)
Board size	0.00413	0.0111	−0.00651	−0.0124	0.00144	0.264***	0.0314	0.017
	(0.0147)	(−0.0126)	(−0.0246)	(0.0197)	(−0.0165)	(0.0357)	(−0.0207)	(−1.02)
Ownership balance	−0.00125	−0.00691*	−0.0401***	−0.00427	−0.00546	−0.184*	−0.00685	−0.008
	(−0.00468)	(−0.00401)	(−0.00873)	(0.00272)	(−0.00532)	(−0.11)	(0.00538)	(−1.58)
Listing age	0.0133	0.0192**	0.0655	0.0105	0.0137	0.0328	0.0312**	0.005
	(0.00944)	(−0.00808)	(0.0423)	(−0.0176)	(−0.0106)	(0.0409)	(−0.014)	(0.51)
Firm size	0.00224	0.0249***	0.0533***	0.00208	0.0293***	−0.093	0.0310***	0.004
	(0.003)	(−0.0026)	(−0.00642)	(−0.0126)	(−0.00543)	(−0.0719)	(−0.00876)	(−1.13)
Leverage	−0.0199	−0.0257*	−0.137***	−0.0314	−0.0863***	0.195***	−0.0434**	−0.027
	(−0.0172)	(−0.0148)	(−0.0262)	(0.0412)	(−0.0197)	(−0.0249)	(−0.02)	(−1.44)
Positive newspaper reports						0.593***		
						(−0.131)		
Positive media coverage						0.417***		
						(−0.0349)		
Constant	6.021***	6.545***	5.247***	6.003***	7.479***	−7.300***	−0.411*	5.930***
	(0.0731)	(0.0632)	(0.188)	(0.161)	(0.0705)	(0.596)	(0.211)	(70.87)
Year fixed effect	yes	yes	yes	yes	yes	yes	yes	yes
Industry fixed effect	yes	yes			yes	yes	yes	yes
Firm fixed effect			yes					
City × industry fixed effect				yes				
Add omitted variables					yes			
Observations	36,000	35,790	35,831	36,000	31,819	33,799	34,416	26,834
R^2	0.958	0.969	0.969	0.959	0.956	0.108	0.952	0.96

Note: Robust standard errors are reported in parentheses (except in Column 4). ***, **, and * denote significance at the 1 %, 5 %, and 10 % levels, respectively.

standard errors at the industry level. As shown in Columns (3) and (4), the coefficient on the linear term of AI washing remains positive (0.0816 and 0.103, respectively), while the coefficient on the quadratic term remains negative (−0.00627 and − 0.0193), further confirming the robustness of the inverted U-shaped effect.

4.3.4. Controlling for omitted variables

To address potential endogeneity arising from omitted variables, we incorporate additional factors that may affect firms' technological gaps: (a) industry competition (Zu and Zhang, 2023), measured by the Lerner Index, where values closer to 1 indicate higher monopoly power and values closer to 0 suggest more competitive markets; (b) managerial capability (Hua et al., 2020), proxied by the natural logarithm of administrative expenses; and (c) financial constraints (Tingbani et al., 2023), measured by the absolute value of the SA index, namely $|-0.737 \times \text{Size} + 0.043 \times \text{Size}^2 - 0.040 \times \text{Age}|$. As reported in Column (5), the linear (β_1=0.115) and quadratic (β_2= − 0.0204) terms of AI washing remain highly significant and consistent with the baseline results, indicating that the non-linear relationship is not driven by omitted variables.

4.4. Endogeneity test

To further mitigate potential endogeneity concerns, this study employs both the two-stage least squares (2SLS) method and the propensity score matching (PSM) approach for endogeneity testing.

4.4.1. 2SLS

To address the potential reverse causality between AI washing and firms' technological capabilities, this study constructs corporate social image as an instrumental variable (IV). Specifically, we use the natural logarithms of the frequency of positive media coverage and positive newspaper reports on listed firms, respectively, derived from the China Financial News Database (CFND), as proxies. The CFND is the first comprehensive database in China that utilizes big data and AI technologies to collect, organize, and analyze financial news related to listed firms in new media. It employs a support vector machine (SVM) model to classify text sentiment, with a high recognition rate for positive and negative tones, enabling it to better capture the media sentiment surrounding listed firms and thus reflect their social image.

This IV satisfies both the relevance and exogeneity conditions: social image is closely tied to AI washing disclosure through firms' communication strategies (Jia and Zhang, 2015), but it is unlikely to directly influence a firm's technological gap. As shown in Column (6) of Table 6, both IVs are significantly positive at the 1 % level in the first-stage regression, indicating a strong positive correlation between corporate social image and AI washing. In the second-stage regression, as shown in Column (7), the linear term of AI washing remains significantly positive (β_1=0.356), and the squared term remains significantly negative (β_2= − 0.0588), consistent with the baseline regression results. Moreover, the IVs pass the weak instrument test (Wald F-statistic = 35.6019 > 10), eliminating the concern of weak instruments. The endogeneity test also supports the validity of the instruments (p = 0.0002), confirming their suitability. Overall, these results confirm that the inverted U-shaped relationship between AI washing, and firms' technological gap remains robust after addressing endogeneity concerns.

4.4.2. Propensity score matching (PSM)

In addition, we employ the PSM method. Firms with AI washing levels above the sample median are assigned to the treatment group, while those below the median are assigned to the control group. Using the nearest neighbor matching method, we match firms and then re-estimate the regression model on the matched sample (see Appendix Table A2, Fig. A1, and Fig. A2 for detailed PSM results). As shown in Column (8) of Table 6, the linear coefficient of AI washing remains significantly positive (β_1=0.059), and the squared term remains significantly negative (β_2= − 0.008), both at the 1 % level. These results again confirm the inverted U-shaped relationship between AI washing and technological gap.

5. Mechanism analysis

5.1. Mediation effect analysis

To identify the mediating mechanism through which AI washing influences firms' technology gap, this study adopts the three-step regression method proposed by Baron and Kenny (1986) to examine the mediating roles of firm-level R&D investment and industry-level R&D investment. The relevant regression results are presented in Table 7.

5.1.1. Firm-level R&D investment

The degree of AI washing is used as the independent variable, and the natural logarithm of the firm's annual R&D investment is taken as the dependent variable to examine its impact on firm-level R&D investment. As shown in Column (1) of Table 7, the linear term of AI washing is negative (β_1= − 0.00666), while the quadratic term is positive (β_2=0.0276) and statistically significant at the 1 % level, indicating a significant U-shaped relationship between AI washing and firm-level R&D investment. This suggests that at low levels of AI washing, firms experience a "resource crowding-out effect" on R&D input; however, when AI washing exceeds a certain threshold, the tension of cognitive dissonance motivates behavioral adjustment, which in turn stimulates a rebound in R&D investment.

Next, both AI washing and R&D investment are included in the regression model with the firm's technology gap as the dependent variable. As reported in Column (2) of Table 7, the coefficients of AI washing (β_1=0.0900; β_2= − 0.0189) remain significant and form an inverted U-shaped relationship. Meanwhile, as the mediating variable, namely firm-level R&D investment, shows a significantly negative coefficient of the mediating variable (β_2= − 0.0190, $p < 0.01$) in Column (2), and the quadratic term in Column (1), namely AIwashing^2 is positive (β_2=0.0276), the product of the two coefficients is negative, indicating that Firm-level R&D investment exerts a significant indirect negative effect in this pathway. Specifically, a higher level of AI washing stimulates firms to increase their R&D investment, and greater R&D input helps narrow the technology gap between the corporate and the industry frontier, thereby establishing a significant mediating mechanism. Additionally, a bootstrap test confirms that the confidence interval for the mediating effect is [0.0255, 0.0292] (excluding 0), and the Sobel test yields a p-value of 0.0000, supporting the significance of the mediation effect. We also calculated the proportion of direct and indirect effects in the total effect: the direct effect accounts for 30.85 %, and the indirect effect accounts for 69.14 %. Detailed results are provided in Table 8. Therefore, Hypothesis H2 is supported.

5.1.2. Industry-level R&D investment

To examine whether AI washing indirectly affects a firm's technology gap through its influence on resource allocation at the industry level, this study uses the natural logarithm of the total annual R&D investment of the firm's industry as a proxy for industry-level R&D investment. As shown in Column (3) of Table 7, the linear term coefficient of AI washing is positive (β_1=0.00571) and the quadratic term coefficient is negative (β_2= − 0.00115), both statistically significant at the 10 % level, suggesting an inverted U-shaped relationship between AI washing and industry-level R&D investment. The economic implication is that when AI washing is at a moderate level, exaggerated signaling by firms may mislead peers about the state of technological progress, triggering a "technology arms race" within the industry. However, when AI washing becomes excessive and the signaling effect loses credibility, peers may become immune to such signaling, leading to a rational retreat in overall industry-level R&D investment.

Table 7
Mediating effect results.

Variables	(1)	(2)	(3)	(4)
	Firm-level R&D investment	Technological gap	Industry-level R&D investment	Technological gap
Firm-level R&D investment		−0.0190***		
		(0.00285)		
Industry-level R&D investment				0.137***
				(0.00727)
AI washing	−0.00666	0.0900***	0.00571*	0.103***
	(0.00910)	(0.00435)	(0.00322)	(0.00444)
AIwashing^2	0.0276***	−0.0189***	−0.00115*	−0.0194***
	(0.00170)	(0.000814)	(0.000593)	(0.000819)
ROA	0.619***	−0.261***	−0.0585*	−0.269***
	(0.108)	(0.0514)	(0.0355)	(0.0491)
Fixed assets ratio	−0.567***	−0.108***	−0.0368**	−0.0336*
	(0.0474)	(0.0227)	(0.0145)	(0.0200)
Executive compensation	0.376***	−0.00725	−0.00509	−0.0133***
	(0.0108)	(0.00529)	(0.00345)	(0.00477)
Board size	−0.196***	0.00488	0.0183*	0.00138
	(0.0328)	(0.0157)	(0.0107)	(0.0148)
Ownership balance	0.0383***	−0.00342	−0.00573*	−0.00394
	(0.0102)	(0.00487)	(0.00342)	(0.00472)
Listing age	−0.368***	0.00832	−0.0193***	0.0155
	(0.0210)	(0.0101)	(0.00690)	(0.00953)
Firm size	0.720***	0.0107***	0.0221***	0.000918
	(0.00693)	(0.00389)	(0.00221)	(0.00305)
Leverage	−0.172***	0.0278	−0.0219*	−0.0260
	(0.0401)	(0.0191)	(0.0126)	(0.0174)
Constant	−2.229***	6.764***	24.22***	2.740***
	(0.168)	(0.0805)	(0.0538)	(0.191)
Year fixed effect	yes	yes	yes	yes
Industry fixed effect	yes	yes	yes	yes
Observations	28,153	28,151	36,151	36,149
R^2	0.574	0.945	0.981	0.957

Note: Robust standard errors are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 8
Mediation test and effect decomposition results.

	Firm-level R&D investment	Industry-level R&D investment
Bootstrap 95 % CI	0.0255–0.0292)	0.0078–0.0164)
Sobel test	$p = 0.00004$	p = 0.0000
Indirect effect	0.012169.14 %)	0.027348.92 %)
Direct effect	0.005430.85 %)	0.028551.08 %)
Total effect	0.0175100 %)	0.0558100 %)

In the full model including the mediating variable, Column (4) of Table 7 shows that AI washing still exhibits a significant inverted U-shaped relationship with the technology gap. Furthermore, the coefficient of industry-level R&D investment is significantly positive (β_2=0.137, $p < 0.01$), and the quadratic term of AIwashing in Column (3) remains negative ($\beta_1 = -0.00115$). Since their product is negative, this suggests that AI washing reduces the technology gap by decreasing industry-level R&D investment. In addition, the bootstrap test shows a confidence interval of [0.0078, 0.0164] (excluding 0), and the Sobel test yields a p-value of 0.0000, further confirming the significance of the industry-level R&D investment as a mediating channel. The decomposition of total effects shows that the direct effect accounts for 48.92 %, while the indirect effect accounts for 51.08 %. Detailed results are provided in Table 8. Therefore, Hypothesis H3 is supported.

5.2. Moderating effect analysis

To further explore the moderating role of learning mechanisms in the relationship between AI washing and corporate technological gap, this study introduces the variable AI-enabled participatory learning capability and examines its moderating effect within the non-linear relationship. The regression results are presented in Table 9.

The findings indicate that the coefficient for the linear term of AI washing is significantly positive (0.101), while the coefficient for the

Table 9
Moderating effect results.

Variables	(1)
AI washing	0.0957***
	(0.00447)
AIwashing^2	−0.0210***
	(0.000830)
AI-enabled participatory learning capability × AI washing	−0.325***
	(0.113)
AI-enabled participatory learning capability × AIwashing^2	0.165***
	(0.0254)
AI-enabled participatory learning capability	0.295*
	(0.178)
ROA	−0.281***
	(0.0514)
Fixed assets ratio	−0.0893***
	(0.0227)
Executive compensation	−0.0142***
	(0.00517)
Board size	0.0100
	(0.0157)
Ownership balance	−0.00432
	(0.00486)
Listing age	0.0163
	(0.0100)
Firm size	−0.00131
	(0.00332)
Leverage	0.0326*
	(0.0191)
Constant	6.758***
	(0.0807)
Year fixed effect	yes
Industry fixed effect	yes
Observations	28,140
R^2	0.945

Note: Robust standard errors are reported in parentheses. ***, **, and * denote significance at the 1 %, 5 %, and 10 % levels, respectively.

quadratic term is significantly negative (−0.0220), both at the 1 % level, confirming the inverted U-shaped relationship between AI washing and the corporate technological gap. More importantly, the interaction term between the quadratic term of AI washing and AI-enabled participatory learning level is significantly positive (0.207, $p < 0.01$), which is opposite in sign to the baseline model. This suggests that firms with higher levels of AI-enabled participative learning capability experience a weakened curvature in the inverted U-shaped relationship, indicating that the technological gap responds less sensitively to variations in AI washing, i.e., the curve flattens. From an economic perspective, at a low level of AI washing, an AI-enabled participatory learning mechanism helps employees understand and absorb AI technologies, reduce the risk of improper resource allocation due to exaggeration, and thus alleviate the expansion of the technological gap. Conversely, when AI washing exceeds a critical threshold, firms with strong AI-enabled participatory learning capability can identify cognitive dissonance and initiate strategic correction earlier, to actively narrow the technological gap through internal capability restructuring.

To further validate the results, we divide the sample into high and low groups based on the median value of AI-enabled participatory learning capability, following Haans et al. (2016). We then compare the curvature steepness of the inverted U-shaped relationship under different levels of AI-enabled participatory learning capability. As shown in Fig. 2, firms with high levels of AI-enabled participatory learning capability exhibit a notably flatter curve compared to those with low levels. This further confirms that the intensity AI washing's impact on the technological gap is significantly moderated by the firm's AI-enabled participatory learning capability. Hypothesis H4 is therefore supported.

6. Heterogeneity analysis

To further explore the boundary conditions under which AI washing influences technological gaps, this study draws upon cognitive dissonance theory to identify three key sources of heterogeneity: investor sentiment, institutional environment, and firm-level technological intensity. These factors may reshape the non-linear relationship between AI washing and technological gap by influencing the tension threshold or adjustment pathway when firms encounter a mismatch between cognition and behavior.

6.1. Investor sentiment

Within the framework of cognitive dissonance theory, a discrepancy between a firm's claims (e.g., AI strategy declarations) and actual behavior (e.g., AI patent accumulation) triggers internal tension, prompting behavioral adjustments. However, the threshold at which this tension activates is not fixed and can be moderated by external market conditions. Investor sentiment, as a reflection of external perceptions, significantly affects firms' tolerance for cognitive dissonance.

Following Rhodes-Kropf et al. (2005), we construct a firm-level investor sentiment index to assess how sentiment environments modulate the non-linear impact of AI washing. Specifically, the firm's market valuation (average Q) is decomposed into an intrinsic value component reflecting growth opportunities and a mispricing component driven by sentiment. To control industry heterogeneity and market cycles, we perform cross-sectional regressions by industry and year as follows:

$$Q_{it} = \beta_0 + \beta_1 \text{firm size}_{it} + \beta_2 \text{leverage}_{it} + \beta_3 \text{ROA}_{it} + \varepsilon_{it} \quad (8)$$

whereas the fitting value of the above regression $Q^f_{i,t}$ is used as a proxy for investment opportunity, while the standardized residual, i.e., $Q^e_{i,t} = Q_{i,t} - Q^f_{i,t}$ (Z-score), is used as the proxy for investor sentiment. The higher the value, the stronger the market sentiment toward the firm.

As shown in Table 10, Column (1), AI washing continues to show a significant inverted U-shaped relationship with the technological gap, with a positive linear term (0.0946) and a negative quadratic term (−0.0169), both of which are significant at the 1 % level. Importantly, the interaction term between the AI washing quadratic term and investor sentiment is significantly negative (−0.00888, $p < 0.01$), suggesting that the higher investor sentiment, the steeper the inverted U-shaped curve. From an economic perspective, the market environment with high investor sentiment sends an optimistic signal to firms and enhances their short-term returns. This increases firms' tolerance for discomfort caused by cognitive dissonance, thereby delaying the trigger point of behavior correction. As a result, the onset of strategic adjustments to AI washing is postponed, leading to more pronounced early-stage divergence followed by stronger catch-up behavior. A visual representation of this heterogeneity effect is provided in Fig. 3.

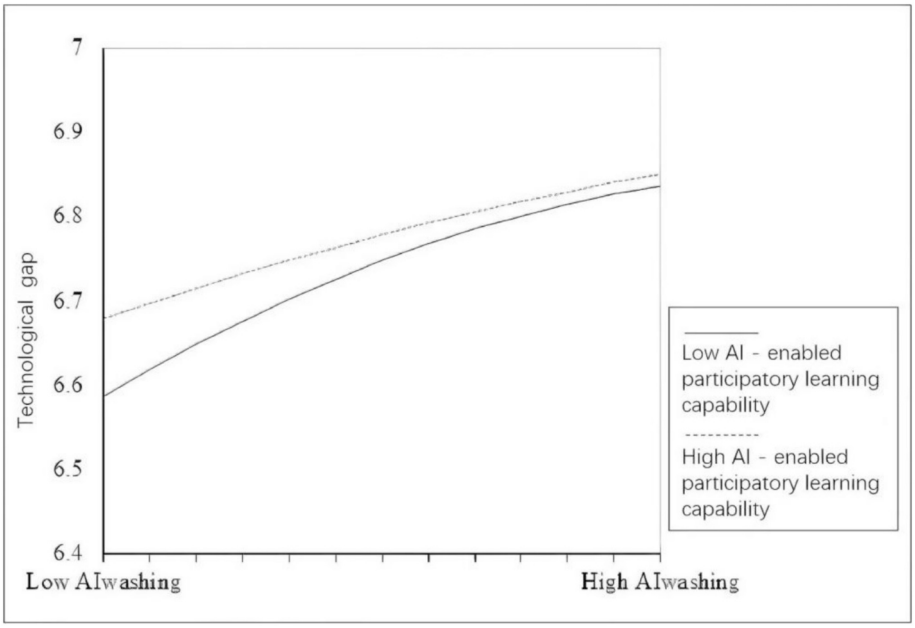


Fig. 2. Moderating effect of AI-enabled participatory learning capability on the relationship between AI washing and technological gap.

Table 10
Heterogeneity analysis.

Variables	(1)	(2)	(3)
AI washing	0.0946***	0.0614***	0.0248***
	(0.00458)	(0.00486)	(0.00650)
AIwashing^2	−0.0169***	−0.0190***	−0.0187***
	(0.000866)	(0.000897)	(0.000809)
Investor sentiment × AI washing	0.0317***		
	(0.00452)		
Investor sentiment × AIwashing^2	−0.00888***		
	(0.000906)		
Investor sentiment	−0.000631		
	(0.00284)		
AI pilot zone × AI washing		0.136***	
		(0.0106)	
AI pilot zone × AIwashing^2		0.00672***	
		(0.00218)	
AI pilot zone		0.0929***	
		(0.00993)	
R&D personnel proportion × AI washing			0.000680***
			(0.000240)
R&D personnel proportion × AIwashing^2			0.000276***
			(0.000048)
R&D personnel proportion			−0.00117***
			(0.000257)
ROA	−0.271***	−0.189***	−0.0510
	(0.0494)	(0.0488)	(0.0447)
Fixed assets ratio	−0.0427**	−0.00865	−0.0662***
	(0.0201)	(0.0200)	(0.0220)
Executive compensation	−0.0112**	0.00169	−0.00495
	(0.00480)	(0.00438)	(0.00480)
Board size	0.00485	0.0157	0.0104
	(0.0149)	(0.0147)	(0.0143)
Ownership balance	−0.00499	−0.00397	−0.00171
	(0.00475)	(0.00473)	(0.00441)
Listing age	0.0111	−0.00145***	0.000412
	(0.00959)	(0.000418)	(0.00971)
Firm size	0.00389	0.00486	−0.0197***
	(0.00307)	(0.015)	(0.00303)
Leverage	−0.0305*	0.0234	0.0292
	(0.0175)	(0.0164)	(0.0177)
Constant	6.003***	8.789***	7.174***
	(0.0747)	(0.846)	(0.0747)
Chow test	P = 0.0000	P = 0.0000	P = 0.0000
Year fixed effect	yes	yes	yes
Industry fixed effect	yes	yes	yes
Observations	36,198	36,000	20,858
R^2	0.957	0.958	0.961

Note: Robust standard errors are reported in parentheses. The row labeled *Chow test* reports the results of the heterogeneity test between groups. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

6.2. Institutional environment

The institutional environment not only shapes the boundaries of corporate behavior but also influences the intensity and path of cognitive dissonance activation and adjustment (Ng and Yu, 2014). In more regulated institutional settings, firms have less tolerance for cognitive dissonance, which accelerates the onset of tension and prompts earlier behavioral correction.

This study exploits the quasi-natural experiment of China's National-Level Artificial Intelligence Pilot Zones launched between 2019 and 2021 to construct a policy-based treatment variable. As of now, the Chinese government has designated 18 cities as national AI pilot zones.[1] Since Deqing County is a county-level jurisdiction and firm locations could not be accurately matched at this level, it was excluded. The remaining 17 prefecture-level cities were designated as the treatment group, and all listed firms located in non-pilot zone cities were categorized as the control group. Accordingly, we define a treatment group dummy variable equal to 1 for firms in pilot zones, and 0 otherwise.[2] As shown in Column (2) of Table 10. AI washing retains its inverted U-shaped impact on the technological gap within the pilot zones, with the linear term (0.0614) and quadratic term (−0.0190) both statistically significant at the 1 % level. However, the interaction term between the AI washing squared term and the pilot zone dummy is significantly positive (0.0672, $p < 0.01$), indicating a notable flattening of the curve. This finding suggests that within national AI pilot zones, listed firms are more likely to constrain the excessive divergence in technological innovation before the AI washing threshold is reached, and likewise moderate their catch-up behavior after surpassing the threshold. The reasoning is that firms located in AI pilot zones are subject to stricter oversight and operate in an environment with better AI expertise and infrastructure. These conditions dampen the incentives for firms to engage in AI washing, thereby softening the non-linearity of its impact. A visual representation of this heterogeneity effect is presented in Fig. 4.

6.3. Technological intensity

The internal technological structure of a firm fundamentally shapes its sensitivity to cognitive dissonance. Compared with other firms, technology-intensive firms typically have a significantly different workforce composition, characterized by a higher proportion of R&D personnel and more mature internal knowledge systems. As a result, the tension between cognition and action induced by AI washing may be more rapidly recognized and addressed, shifting the correction pathway forward.

Following Souitaris (2002), we classify firms based on the proportion of R&D personnel in their workforce. Firms above the median are categorized as technology-intensive and assigned a dummy variable of 1; those below are non-technology-intensive and assigned a value of 0. As shown in Column (3) of Table 10, in technology-intensive firms, the relationship between AI washing and the technological gap is inverted U-shaped, with a linear coefficient of 0.0248 and a quadratic coefficient of −0.0187 (both significant at the 1 % level). Moreover, the interaction term between AI washing square sum and the technological intensity dummy is significantly positive (0.000276), indicating low curvature. This finding implies that in technology-intensive firms, the tendency to suppress the excessive divergence of technological gaps is stronger before reaching the turning point. Similarly, after exceeding the threshold, these firms are also more likely to moderate their catch-up efforts. There are two main reasons for this. First, their strong technological foundation plays a buffer role, which can resist the divergence effects of AI washing; Second, their catch-up strategies tend to be more selective and consistent with long-term strategic goals, rather than blind imitation. Therefore, the inverted U-shaped relationship is significantly flattened. A visual representation of the heterogeneity effect of technological intensity can be found in Fig. 5.

7. Discussion and conclusion

This study employs the Cognitive Dissonance Theory to empirically examine the impact of corporate 'AI washing' on the technological gap, thereby addressing critical research gaps concerning the lack of quantitative measurement (Bietti, 2020) and empirical evidence on technological effects in existing literature (Seele and Schultz, 2022).

The primary finding reveals an inverted U-shaped relationship

[1] List of cities in order of designation: Beijing, Shanghai, Tianjin, Shenzhen, Hangzhou, Hefei, Deqing County (excluded), Chongqing, Chengdu, Xi'an, Jinan, Guangzhou, Wuhan, Suzhou, Changsha, Zhengzhou, Shenyang, and Harbin.

[2] To account for the implementation lag following policy announcements, if a city was designated after September of a given year, the treatment variable is coded as 1 starting from the following year.

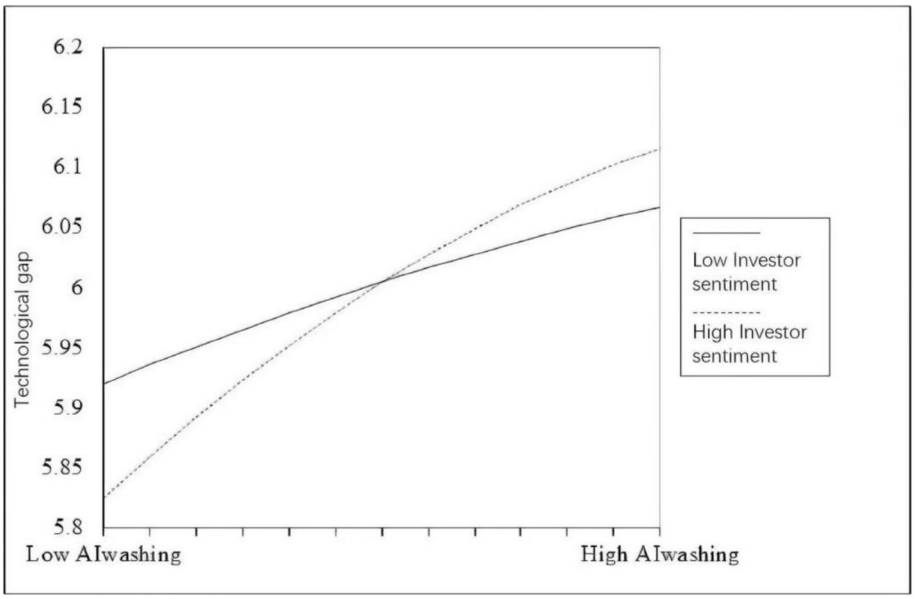


Fig. 3. Heterogeneity effect of investor sentiment.

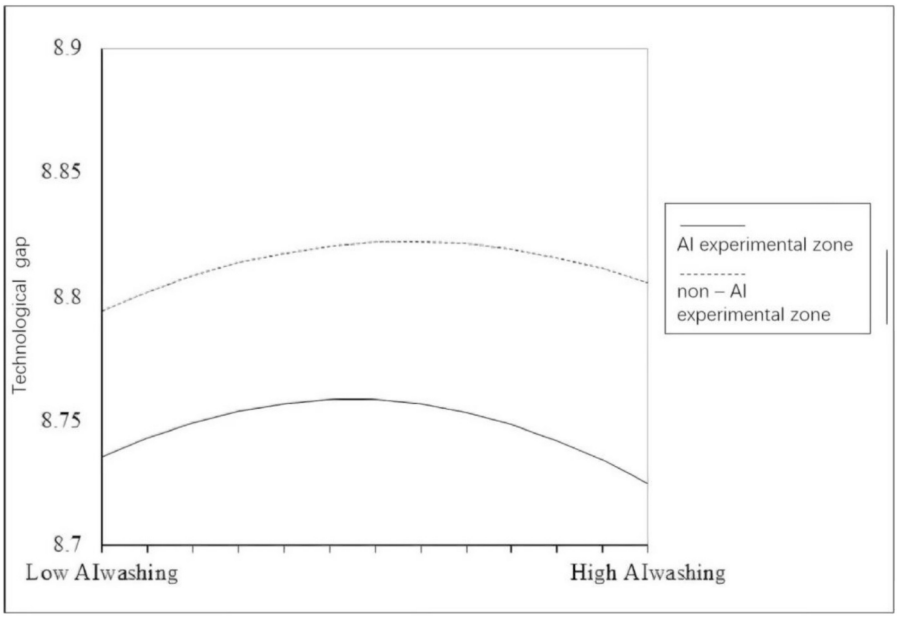


Fig. 4. Heterogeneity effect of institutional environment.

between corporate 'AI washing' and the technological gap. This challenges the prevailing linear assumption that 'AI washing' uniformly hinders technological innovation (Seele and Schultz, 2022), offering a nonlinear extension of Cognitive Dissonance Theory within the context of corporate technology strategy. This relationship is transmitted through a dual-path mechanism involving both firm- and industry-level R&D investment, which moves beyond a static analytical framework. Specifically, at the firm level, 'AI washing' initially crowds out R&D resources (Adomako and Tran, 2023), but the accumulation of cognitive dissonance eventually motivates firms to increase R&D investment to restore consistency, confirming the theory's core tenet. At the industry level, while 'AI washing' intensifies market competition (Carrier and Tushnet, 2021), our analysis reveals a dynamic process where industry-level R&D investment first rises and subsequently returns to a more reasonable level, thus refining the logical framework of industrial spillover effects.

Furthermore, moderating effect analysis indicates that firms with stronger AI-enabled participatory learning capability exhibit a smoother inverted U-shaped adjustment and a lower cognitive dissonance threshold, extending research on organizational learning. Also, heterogeneity analysis further deepens understanding of strategic decision-making: high investor sentiment delays the adjustment threshold, confirming the role of market sentiment (Basnet et al., 2025). Moreover, a well-established institutional environment is shown to effectively reduce firms' tolerance for cognitive dissonance, providing empirical support for the constraining role of institutional pressure (Jobin et al., 2019). Finally, technology-intensive firms exhibit greater sensitivity to cognitive dissonance, aligning with discussions on strategic response variations (Shi, 2024) and offering an important theoretical basis for explaining differences in corporate behavioral adjustments.

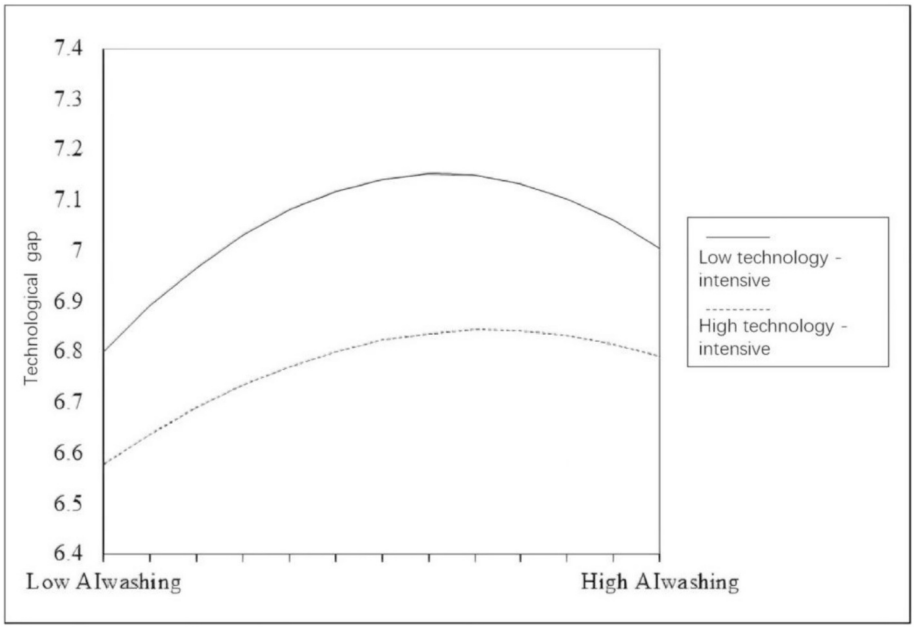


Fig. 5. Heterogeneity effect of technological intensity.

7.1. Theoretical implications

This research makes three key theoretical contributions. First, this study promotes the leap of AI washing research from conceptualization to empirical analysis, expands the theoretical boundary of this field, and empirically tests the behavioral consequences and dynamic evolution logic of AI washing. Furthermore, this study reveals the correction mechanism of AI washing, that is, compensatory R&D behavior triggered by cognitive dissonance, which forms a different dialogue from the traditional signaling theory. Proposed by Spence (1973), the signaling theory emphasizes the stable transmission and effectiveness of real signals, but this study finds that AI washing, a false technological signal, does not remain valid indefinitely. When the deviation accumulation exceeds the organizational tolerance threshold, behavior adjustments will be activated, such as increasing R&D investment. This finding shifts AI washing research from an ethical issue to organizational behavior analysis and provides a new perspective of the interaction between cognition and behavior for AI washing research.

Second, this paper extends cognitive dissonance theory to the field of AI strategy, enriching the theoretical explanation of organizational level responses to dissonance. Proposed by Festinger (1957), cognitive dissonance theory was initially used to explain conflicts between cognition and behavior at the individual level; although some studies have extended it to the organizational level, none have involved the emerging AI strategic context. This study defines AI washing as a typical scenario of organizational cognitive dissonance, providing a theoretical foundation for understanding the emergence and evolution of AI washing. Additionally, this paper identifies the moderating role of AI-enabled participatory learning capability in dissonance resolution. This finding integrates cognitive dissonance theory with AI technology characteristics, expands the application boundary of the theory in the field of technology driven organizational behavior, and addresses the limitation of traditional organizational dissonance research that ignores the moderating role of technological capabilities.

Finally, this study reconstructs the evolution logic of technological gaps and supplements the theoretical perspective of cognition-driven dynamic adjustment. Traditional research on technological gaps mostly focuses on structural factors. For instance, the external level emphasizes the impact of institutional environments (Ng and Yu, 2014) and market competition (Zu and Zhang, 2023), while the internal level focuses on resource endowments such as human capital (Burger et al., 2022) and R&D investment (Fuenfschilling and Truffer, 2014). Moreover, the existing studies usually regard technological gap as the differences in terms of static capability (Bourles et al., 2013), while insufficient attention is paid to the driving role of cognitive factors. Based on cognitive dissonance theory, this study proposes a new logic. The evolution of technological gaps is a dynamic process. In this process, cognitive dissonance affects resource allocation adjustments and leads to changes in the technological gap. This nonlinear logic breaks the traditional idea and provides a richer analytical framework for technological gap research.

In terms of theoretical consistency, the conclusion that AI washing widens the technological gap in the early stage is highly consistent with the classic view in greenwashing literature that symbolic behavior squeezes out substantive innovation investment. Delmas and Burbano (2011) pointed out that when firms carry out greenwashing by exaggerating environmental commitments, they will transfer resources from environmental technology R&D to public relations and marketing, which weakens their sustainable innovation ability. Similarly, Seele and Schultz (2022) proposed that firms' false assertion of AI capabilities will lead to a resource mismatch effect, giving priority to technology description over actual R&D. Our study further confirms through quantitative data that this mismatch is manifested in the dynamic "squeeze and rebound" mode of R&D investment, which timely responds to the conclusions of the above literature.

7.2. Practical implications

The findings of this study offer several actionable implications for various practitioners. For regulatory authorities, it is essential to improve the regulatory framework for AI-related disclosure and strengthen substantive review mechanisms. Specifically, regulators must formulate differentiated AI disclosure standards based on industry-specific characteristics. In technology-intensive sectors, disclosures should be cross-validated against objective indicators such as patent quality and technology commercialization outcomes to enhance the regulatory accuracy concerning 'AI washing' behavior. Concurrently, legislative bodies should accelerate the development of AI-specific regulations that explicitly specify the legal liabilities and penalties associated with 'AI washing'.

From an industry perspective, industry associations should adopt a proactive role in mitigating information asymmetry within the AI domain. They can collaborate with research institutions to develop a multi-dimensional and quantifiable evaluation system for assessing actual corporate AI capabilities. Such a system would establish an objective basis for informed regulatory actions and corporate decision-making, thereby countering the market distortions caused by 'AI washing'. Furthermore, industry associations should promote the establishment of platforms for R&D resources and information sharing. This would help prevent inefficient R&D competition triggered by 'AI washing', ultimately contributing to the optimal allocation of innovative resources.

For firms, decision-makers must construct a robust internal safeguard against 'AI washing'. On one hand, firms should increase investment in AI-related human capital and cultivate high-performing teams skilled in human-computer collaboration. On the other hand, they must strengthen the governance of AI strategy and avoid overhyping AI concepts solely to cater to capital markets. Instead, resources should be strategically directed toward substantive technological innovation to ensure that AI strategies generate sustainable long-term value.

7.3. Limitations and directions for future research

This study possesses several limitations that point to promising avenues for future research.

First, the research sample is drawn exclusively from Chinese listed firms. While this provided a necessary representative context for this study, it does not fully capture the impact of 'AI washing' on the technological gap dynamics of small and medium-sized firms (SMEs) and non-listed enterprises. Future studies could significantly broaden the sample scope (e.g., cross-country analysis) to examine how the cognitive dissonance mechanism operates heterogeneously across diverse firm types and institutional environments, enhancing the generalizability of the findings.

Second, the measurement of 'AI washing' is primarily based on the discrepancy between textual disclosures and patent data. This approach, while standard, does not fully account for the quality of AI innovation or its practical application success. Future research should incorporate multidimensional indicators such as the success rate of AI model deployment, AI-related trademarks, or quantifiable measures of project implementation success to substantially improve the accuracy and robustness of the 'AI washing' measure.

Third, although this study identifies a nonlinear inflection point and heterogeneity in corporate behavioral adjustments, it does not fully trace the dynamic evolutionary trajectory of these adjustments over time. Going forward, researchers could employ advanced methods such as event studies or sophisticated longitudinal panel models to more comprehensively capture the entire process through which 'AI washing' induces cognitive dissonance and ultimately affects the corporate technological gap.

CRediT authorship contribution statement

Zhe Sun: Writing – original draft, Methodology, Data curation, Conceptualization. **Yujun Wen:** Writing – original draft, Visualization, Software, Methodology, Formal analysis. **Liang Zhao:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Intesar Almugren:** Writing – original draft, Validation, Supervision, Resources, Project administration, Funding acquisition. **Aradhana Galgotia:** Validation, Resources, Project administration, Investigation.

Acknowledgments

Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2025R429), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

Appendix A

Table A1
AI lexicon.

AI	AI Products	AI Chips	Machine Translation	Machine Learning
Computer Vision	Human-Computer Interaction	Deep Learning	Neural Networks	Biometrics
Image Recognition	Data Mining	Feature Recognition	Speech Synthesis	Speech Recognition
Knowledge Graphs	Smart Banking	Intelligent Insurance	Human-Machine Collaboration	Intelligent Regulation
Smart Education	Intelligent Customer Service	Smart Retail	Smart Agriculture	Robo-Advisors
Augmented Reality	Virtual Reality	Smart Healthcare	Smart Speakers	Intelligent Voice
Smart Governance	Autonomous Driving	Intelligent Transportation	Convolutional Neural Networks	Voiceprint Recognition
Feature Extraction	Driverless Cars	Smart Home	Question-Answering Systems	Facial Recognition
Business Intelligence	Smart Finance	Recurrent Neural Networks	Reinforcement Learning	Intelligent Agents
Elderly Care AI	Big Data Marketing	Big Data Risk Control	Big Data Analytics	Big Data Processing
Support Vector Machines (SVM)	Long Short-Term Memory (LSTM)	Robotic Process Automation	Natural Language Processing	Distributed Computing
Knowledge Representation	Intelligent Chips	Wearable Devices	Big Data Management	Smart Sensors
Pattern Recognition	Edge Computing	Big Data Platforms	Intelligent Computing	Intelligent Search
Internet of Things	Cloud Computing	Augmented Intelligence	Voice Interaction	Smart Environmental Protection
Human-Machine Dialogue	Deep Neural Networks	Big Data Operations		

Note: This lexicon of 73 AI-related terms was manually curated based on the Chinese translation of AI-related vocabulary provided by Chen and Srinivasan (2024), Ping An Securities' AI Industry Chain Panorama Report on the science and technology innovation board (STAR Market), the 2019 China AI Industry Market Outlook Report by China Commercial Industry Research Institute, the 2019 AI Industry Status and Development Trends Report by Shenzhen Qianzhan Industry Research Institute, and the AI glossary published by the World Intellectual Property Organization (WIPO). Initially, 52 terms were selected as seed words. Subsequently, the corpus used for training included vocabulary extracted from annual reports and patent documents. For each seed word, the ten most semantically similar terms were identified based on cosine similarity. Redundant terms, words irrelevant to AI, and those with excessively low frequency were removed, resulting in a final lexicon of 73 terms. It is worth noting that when calculating the frequency of terms such as Support Vector Machine (SVM), the frequency is defined as the sum of occurrences of both the Chinese and English versions of the term. Additionally, abbreviations that may lead to ambiguity, such as CNN (which could also refer to the Cable News Network, www.cnn.com), were excluded from the lexicon to avoid misinterpretation. Finally, terms such as Neural Networks and Convolutional Neural Networks are counted independently. This is because frequency statistics were based on tokenized text rather than raw documents; thus, after tokenization, these terms are treated as separate entities.

Table A2
PSM results.

Variable	Unmatched Matched	Mean Treated	Mean Control	% reduction %bias	% reduction |bias|	t—test t	t—test p > |t|	V(T) V(C)
ROA	U	0.03922	0.03539	6		5.7	0	1.13*
	M	0.03921	0.03833	1.4	77	1.27	0.204	1.06*
Fixed assets ratio	U	0.23764	0.20467	20.4		19.27	0	0.60*
	M	0.23753	0.24028	−1.7	91.6	−1.55	0.122	0.54*
Executive compensation	U	15.145	15.369	−28.5		−27.06	0	0.91*
	M	15.145	15.153	−0.9	96.8	−0.88	0.379	1
Board size	U	2.1161	2.1504	−17.3		−16.39	0	0.87*
	M	2.1162	2.1205	−2.1	87.5	−2.02	0.044	0.88*
Ownership balance	U	0.73064	0.7091	3.6		3.39	0.001	0.97
	M	0.73068	0.73113	−0.1	97.9	−0.07	0.945	0.93*
Listing age	U	2.8616	2.8922	−8.8		−8.35	0	1.01
	M	2.8616	2.8592	0.7	92	0.65	0.513	1.01
Firm size	U	21.892	22.46	−44.8		−42.42	0	0.62*
	M	21.893	21.898	−0.4	99	−0.44	0.658	0.87*
Leverage	U	0.40075	0.46399	−30.7		−29.2	0	0.90*
	M	0.40084	0.39975	0.5	98.3	0.5	0.614	0.97

Note: This table reports the matching results of the PSM method.

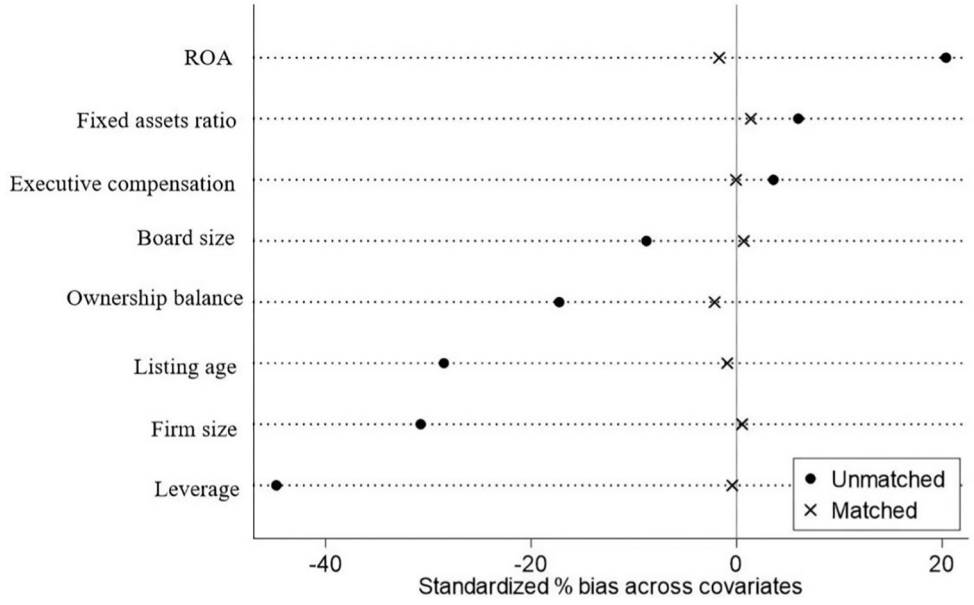


Fig. A1. Standardized bias.

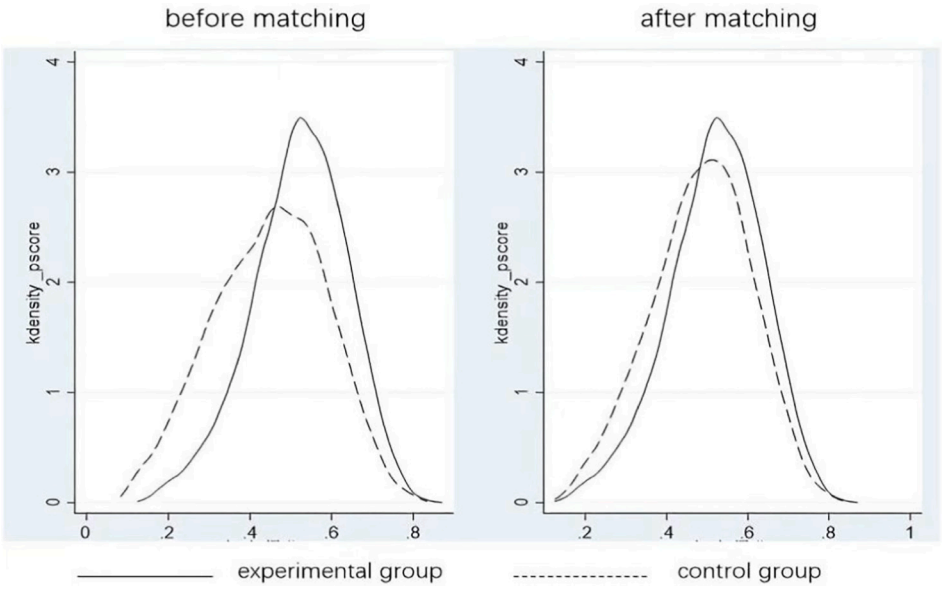


Fig. A2. Kernel density estimation.

Data availability

Data will be made available on request.

References

Adomako, S., Tran, M.D., 2023. Doing well and being responsible: the impact of corporate social responsibility legitimacy on responsible entrepreneurship. Corp. Soc. Responsib. Environ. Manag. 30 (4), 1794–1804. https://doi.org/10.1002/csr.2455.

Ali, A.I., Lerme, C.S., 1997. Comparative advantage and disadvantage in DEA. Ann. Oper. Res. 73, 215–232. https://doi.org/10.1023/a:1018929228294.

Auster, D., 1965. Attitude-change and cognitive-dissonance. J. Mark. Res. 2 (4), 401–405. https://doi.org/10.2307/3149488.

Bahoo, S., Cucculelli, M., Qamar, D., 2023. Artificial intelligence and corporate innovation: a review and research agenda. Technol. Forecast. Soc. Chang. 188, 122264. https://doi.org/10.1016/j.techfore.2022.122264.

Bai, S., Tan, Y., Han, C., Yang, M., Gupta, B.B., Arya, V., Kumar, N., 2024. Unveiling the evolution of enterprise digital innovation strategies: insights from us-listed companies' annual reports. Ieee Transactions on Computational Social Systems. 11 (6), 8243–8256. https://doi.org/10.1109/tcss.2024.3360040.

Baron, R.M., Kenny, D.A., 1986. The moderator–mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations. J. Pers. Soc. Psychol. 51 (6), 1173.

Basnet, A., Elias, M., Salganik-Shoshan, G., Walker, T., Zhao, Y., 2025. Analyzing the market's reaction to AI narratives in corporate filings [article]. Int. Rev. Financ. Anal. 105, 104378.

Bhimani, H., Mention, A.L., Salampasis, D., 2023. Disengagement in open innovation: a cognitive perspective. Br. J. Manag. 34 (1), 241–258. https://doi.org/10.1111/1467-8551.12594.

Bietti, E., 2020. From ethics washing to ethics bashing: a view on tech ethics from within moral philosophy. In: Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency, pp. 210–219.

Bourles, R., Cette, G., Lopez, J., Mairesse, J., Nicoletti, G., 2013. Do product market regulations in upstream sectors curb productivity growth? Panel data evidence for oecd countries. Rev. Econ. Stat. 95 (5), 1750–1768. https://doi.org/10.1162/REST_a_00338.

Burger, M.J., Kounetas, K., Napolitano, O., Stavropoulos, S., 2022. Do innovation and human capital actually narrow the technology gap? Champions and laggards of European regional productive performance. Reg. Stud. 56 (10), 1655–1670. https://doi.org/10.1080/00343404.2021.2000596.

Carrier, M.A., Tushnet, R., 2021. An antitrust framework for false advertising. Iowa Law Review. 106 (4), 1841–1884.

Chen, W., Srinivasan, S., 2024. Going digital: implications for firm value and performance. Rev. Acc. Stud. 29 (2), 1619–1665. https://doi.org/10.1007/s11142-023-09753-0.

Coad, A., 2019. Persistent heterogeneity of RandD intensities within sectors: evidence and policy implications. Res. Policy 48 (1), 37–50. https://doi.org/10.1016/j.respol.2018.07.018.

Delmas, M.A., Burbano, V.C., 2011. The drivers of greenwashing. Calif. Manag. Rev. 54 (1), 64–74. https://doi.org/10.1525/cmr.2011.54.1.64.

Denia, E., 2025. AI narratives model: social perception of artificial intelligence. Technovation 146, 103266.

Elsayed, N., 2025. Strategic AI disclosures and legitimacy: impression management in UK FTSE100 annual reports. Int. J. Account. Inf. Syst. 56, 100757. https://doi.org/10.1016/j.accinf.2025.100757.

Falco, G., 2019. Participatory AI. In 2019 22nd IEEE International Conference on Computational Science and Engineering (CSE) and 17th IEEE International Conference on Embedded and Ubiquitous Computing (EUC). IEEE.

Flagstad, I., Hauge, A.L., Johnsen, S.A.K., 2022. Certification dissonance: contradictions between environmental values and certification scheme requirements in small-scale companies. J. Clean. Prod. 358, 132037. https://doi.org/10.1016/j.jclepro.2022.132037.

Fuenfschilling, L., Truffer, B., 2014. The structuration of socio-technical regimes-conceptual foundations from institutional theory. Res. Policy 43 (4), 772–791. https://doi.org/10.1016/j.respol.2013.10.010.

Haans, R.F., Pieters, C., He, Z.L., 2016. Thinking about U: Theorizing and testing U-and inverted U-shaped relationships in strategy research. Strateg. Manag. J. 37 (7), 1177–1195.

Hua, N., DeFranco, A., Abbott, J., 2020. Management fees and hotel performance in the US. Tour. Manag. 79, 104093. https://doi.org/10.1016/j.tourman.2020.104093.

Jia, M., Zhang, Z., 2015. News visibility and corporate philanthropic response: evidence from privately owned chinese firms following the wenchuan earthquake. J. Bus. Ethics 129 (1), 93–114. https://doi.org/10.1007/s10551-014-2150-7.

Jobin, A., Ienca, M., Vayena, E., 2019. The global landscape of AI ethics guidelines. Nature Machine Intelligence. 1 (9), 389–399. https://doi.org/10.1038/s42256-019-0088-2.

Kim, S., Bay, D., 2017. Cognitive dissonance as an explanation of goodwill write-offs. J. Behav. Financ. 18 (1), 14–28. https://doi.org/10.1080/15427560.2017.1274755.

Li, Z., Wen, F., Huang, Z.J., 2023. Asymmetric response to earnings news across different sentiment states: the role of cognitive dissonance. Finance 78, 102343. https://doi.org/10.1016/j.jcorpfin.2022.102343.

Liang, Y., Giroud, A., Rygh, A., 2022. Strategic asset-seeking acquisitions, technological gaps, and innovation performance of Chinese multinationals. J. World Bus. 57 (4), 101325. https://doi.org/10.1016/j.jwb.2022.101325.

Liu, Y., Shen, F., Guo, J., Hu, G., Song, Y., 2025. Can artificial intelligence technology improve companies' capacity for green innovation? Evidence from listed companies in China. Energy Econ. 143, 108280. https://doi.org/10.1016/j.eneco.2025.108280.

Ng, T., Yu, L., 2014. Which types of institutions hinder productivity among private manufacturing firms in China? China Econ. Rev. 31, 17–31. https://doi.org/10.1016/j.chieco.2014.07.010.

Powers, T.L., Jack, E.P., 2013. The influence of cognitive dissonance on retail product returns. Psychol. Mark. 30 (8), 724–735. https://doi.org/10.1002/mar.20640.

Pugh, S.D., Groth, M., Hennig-Thurau, T., 2011. Willing and able to fake emotions: a closer examination of the link between emotional dissonance and employee well-being. J. Appl. Psychol. 96 (2), 377–390. https://doi.org/10.1037/a0021395.

Rabin, M., 1994. Cognitive-dissonance and social-change. J. Econ. Behav. Organ. 23 (2), 177–194. https://doi.org/10.1016/0167-2681(94)90066-3.

Rhodes-Kropf, M., Robinson, D.T., Viswanathan, S., 2005. Valuation waves and merger activity: the empirical evidence. J. Financ. Econ. 77 (3), 561–603. https://doi.org/10.1016/j.jfineco.2004.06.015.

Saemundsson, R.J., Candi, M., 2017. Absorptive capacity and the identification of opportunities in new technology-based firms. Technovation 43–49. https://doi.org/10.1016/j.technovation.2017.06.001.
Scher, S.J., Cooper, J., 1989. Motivational basis of dissonance - the singular role of behavioral consequences. J. Pers. Soc. Psychol. 56 (6), 899–906. https://doi.org/10.1037/0022-3514.56.6.899.
Seele, P., Schultz, M.D., 2022. From greenwashing to machinewashing: a model and future directions derived from reasoning by analogy. J. Bus. Ethics 178 (4), 1063–1089. https://doi.org/10.1007/s10551-022-05054-9.
Sheng, Z., Ni, N., 2024. Dynamic game analysis of E-commerce platform rewards and research and development investment of settled enterprises. Int. Rev. Econ. Financ. 92, 1112–1125. https://doi.org/10.1016/j.iref.2024.02.068.
Shi, J., 2024. Adaptive change: emerging economy enterprises respond to the international business environment challenge. Technovation 133, 102998. https://doi.org/10.1016/j.technovation.2024.102998.
Souitaris, V., 2002. Firm-specific competencies determining technological innovation: a survey in Greece. R D Manag. 32 (1), 61–77. https://doi.org/10.1111/1467-9310.00239.
Spence, M., 1973. Job market signaling. Q. J. Econ. 87 (3), 355–374.
Sun, Y., Tao, Q., Wang, D., Zhang, W., 2025b. Corporate ESG decoupling and RandD investment. N. Am. J. Econ. Financ. 75, 102278. https://doi.org/10.1016/j.najef.2024.102278.
Sun, Z., Sun, X., Wang, W., Wang, W., 2025a. Digital transformation and greenwashing in environmental, social, and governance disclosure: does investor attention matter? Business Ethics the Environment and Responsibility. 34 (1), 81–102. https://doi.org/10.1111/beer.12585.
Theodorou, A., Dignum, V., 2020. Towards ethical and socio-legal governance in AI. Nature Machine Intelligence. 2 (1), 10–12. https://doi.org/10.1038/s42256-019-0136-y.
Tingbani, I., Salia, S., Hussain, J.G., Alhassan, Y., 2023. Environmental tax, SME financing constraint, and innovation: evidence from OECD countries. IEEE Trans. Eng. Manag. 70 (3), 1006–1025. https://doi.org/10.1109/tem.2021.3110812.
Wang, S., Huang, X., Xia, M., Shi, X., 2024. Does artificial intelligence promote firms' innovation efficiency: evidence from the robot application. J. Knowl. Econ. 2024 (15), 16373–16394. https://doi.org/10.1007/s13132-023-01707-w.
Wang, T., Cheng, D., 2023. Executive shareholding, institutional investor shareholding and enterprise innovation. Eur. J. Innov. Manag. 26 (6), 1744–1762. https://doi.org/10.1108/ejim-11-2021-0553.
Wilson, C., Atabey, A., Revans, J., 2025. Towards child-centred AI in children's learning futures: participatory design futuring with smartschool and the co-design stories toolkit. International Journal of Human-Computer Studies. 199, 103431. https://doi.org/10.1016/j.ijhcs.2024.103431.
Xing, X., Zhang, Z., He, W., 2026. AI technology, AI narrative, and firm value. Technovation 149, 103349.
Xu, Y., Ji, J., Li, N., Borah, D., 2024. How do executive excess compensation affect enterprise technological innovation: evidence from a panel threshold model of chinese biopharmaceutical companies. J. Bus. Res. 179, 114683. https://doi.org/10.1016/j.jbusres.2024.114683.
Zhang, D., 2022. Are firms motivated to greenwash by financial constraints? Evidence from global firms' data. J. Int. Financ. Manag. Acc. 33 (3), 459–479. https://doi.org/10.1111/jifm.12153.
Zhang, D., 2023. Can environmental monitoring power transition curb corporate greenwashing behavior? J. Econ. Behav. Organ. 212, 199–218. https://doi.org/10.1016/j.jebo.2023.05.034.
Zirar, A., Ali, S.I., Islam, N., 2023. Worker and workplace artificial intelligence (AI) coexistence: emerging themes and research agenda. Technovation 124, 102747. https://doi.org/10.1016/j.technovation.2023.102747.
Zu, Y., Zhang, R., 2023. Supplier change, market competition and enterprise innovation. Eur. J. Innov. Manag. 26 (4), 1034–1053. https://doi.org/10.1108/ejim-09-2021-0480.

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